

EXPERIMENT NO: 3

Heuristic Evaluation of an Existing Website or App - Prarthana D/ 212225230213

AIM:

To evaluate and compare the usability of selected applications (Swiggy Instamart and Blinkit) using heuristic principles and analyze improvements using A/B testing.

OBJECTIVES:

- To understand Nielsen's heuristic evaluation principles.
- To compare the UX of competing quick-commerce applications.
- To identify usability and interaction design issues.
- To suggest actionable design improvements.
- To evaluate the impact of improvements using A/B testing.

ALGORITHM:

1. Select two competing applications (**Swiggy Instamart** and **Blinkit**).
2. Study Nielsen's usability heuristics (visibility of system status, consistency, error prevention, aesthetic and minimalist design, etc.).
3. Analyze both applications based on these principles across critical user flows (search, cart addition, checkout).
4. Identify usability bottlenecks and design flaws in each app.
5. Suggest targeted design improvements.
6. Create two interface versions (**Version A** – original design, **Version B** – improved design).
7. Perform A/B testing to compare user interactions and task success.
8. Record, analyze, and present findings.

OUTPUT

a. Heuristic Evaluation

Heuristics Used:

- Visibility of system status
- Match between system and real world
- User control and freedom
- Consistency and standards
- Error prevention
- Recognition rather than recall

- Aesthetic and minimalist design

b. Comparison of Applications

Feature	Swiggy Instamart	Blinkit
Navigation & Discovery	Integrated within the Swiggy super-app; category hierarchy can feel multi-layered	Dedicated quick-commerce layout; direct, prominent category grid
UI Density & Clutter	Moderately cluttered due to cross-promotions and food delivery banners	High visual density with prominent deals, flash sales, and add-on carousels
Search Functionality	Context-aware with typo tolerance; occasionally mixes restaurant dishes if global search is used	Highly responsive, focused purely on groceries and daily staples
Cart & Add-on Flow	Smooth item counter; subtle suggestions at bottom sheet	Aggressive checkout prompts and cross-sell nudges prior to payment
Speed & Responsiveness	Fast, but transitions depend on the parent super-app container	Very fast, dedicated single-purpose app architecture
Visibility of Status	Clear live driver tracking and minute-by-minute order milestones	Real-time packing and delivery map with explicit delivery countdown

c. Identified Issues

- **Swiggy Instamart:**
 - **Super-app Friction:** The shared search bar occasionally causes cross-category confusion between food delivery and grocery items.
 - **Visual Overload:** Persistent promotional banners and cross-vertical nudges distract from quick repeat purchases.
 - **Cart Modification:** Adjusting item quantities inside deep checkout sheets can require extra taps.
- **Blinkit:**
 - **Aggressive Cross-Selling:** Multiple pop-ups and bottom sheets suggesting last-minute additions create cognitive friction before checkout.
 - **Dense Visual Hierarchy:** Bright discount tags and high-contrast badges can make scanning ingredient lists or pack sizes harder.
 - **Accidental Taps:** Small increment/decrement target areas on product cards lead to occasional misclicks.

d. Proposed Improvements

- **Swiggy Instamart:**
 - Isolate the Instamart search index cleanly when the user is inside the grocery tab.
 - Streamline the home view by reducing redundant promotional carousels.
 - Add a quick-reorder widget directly accessible from the bottom navigation bar.
- **Blinkit:**
 - Consolidate last-minute add-on recommendations into a single, dismissible horizontal tray instead of multiple interstitial prompts.
 - Increase the hit targets and spacing for quantity buttons on product listing cards.
 - Soften secondary badge colors to emphasize primary product details (weight, volume, price).

e. A/B Testing

- **Version A (Original Design):**
 - Existing layout with standard add-on nudges, current card touch targets, and default search behavior.
- **Version B (Improved Design):**
 - Decluttered checkout flow with a single-step add-on tray, isolated search results, and enlarged interactive buttons.
- **User Behavior Observed:**
 - Task completion time (search to final payment screen).
 - Error rate (accidental cart adds and misclicks).
 - Task drop-off rate at the pre-payment stage.
 - User satisfaction score (System Usability Scale).

- **Inference:**

- **Version B** reduced total checkout time by eliminating unnecessary interstitial nudges.
- Larger button touch targets significantly decreased misclicks on product listing cards.
- Overall user satisfaction increased due to a more focused and predictable checkout journey.

RESULT:

Thus, the heuristic evaluation of Swiggy Instamart and Blinkit was successfully conducted, usability bottlenecks were identified, and interface improvements were validated using A/B testing methodology.